

Portfolio Choice and Trading Volume with Loss-Averse Investors

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Abstract

This paper presents a model of portfolio choice and stock trading volume with loss-averse investors. The demand function for risky assets is discontinuous and non-monotonic: as wealth rises beyond a threshold investors follow a generalized portfolio insurance strategy. This behavior is consistent with the evidence in favor of the disposition effect. In addition, loss-averse investors will not hold stocks unless the equity premium is quite high. The elasticity of the aggregate demand curve changes substantially, depending on the distribution of wealth across investors. In an equilibrium setting the model generates positive correlation between trading volume and stock return volatility, but suggests that this relationship should be non-linear.

Key Words: Loss Aversion, First-Order Risk Aversion, Portfolio Insurance, Portfolio Choice, Stock Market Participation, Trading Volume

JEL Classification: G11, G12

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1 Introduction

“Value should be treated as a function in two arguments: the asset position that serves as a reference point, and the magnitude of the change (positive or negative) from that reference point”, Kahneman and Tversky (1979).

This paper solves a model of portfolio choice and trading volume with loss-averse investors. Loss aversion specifies that individuals value wealth relative to a given reference point, that they are (much) more sensitive to losses than to gains (both measured relative to the reference point), and that they are risk-averse in the domain of gains and risk-loving in the domain of (moderate) losses. The first property is summarized in the above quote from Kahneman and Tversky (1979). The second property corresponds to the notion of first-order risk aversion as discussed by Epstein and Zin (1990). It implies that agents exhibit significant risk aversion even for very small gambles. The last property states that following losses the investor is more willing to take additional risks (so as to be able to go back to the break-even point), while following gains she will be more conservative.

If investors exhibit first-order risk aversion and their attitudes towards risk are a function of the past performance of their investments, then this will have important implications for the demand for risky assets and, in equilibrium, for the conditional distribution of stock returns and for trading volume.

We start by studying the optimal portfolio allocation behavior of a loss-averse investor. This behavior depends crucially on the level of surplus wealth (current wealth relative to the reference point), and on how the investor’s reference point reacts to changes in the current stock price. As surplus wealth reaches a certain threshold, the investor sells a significant part of her stock holdings and follows a (generalized) portfolio insurance rule, protecting herself against losses (relative to her reference point). Intuitively, as the stock price goes up, the investor faces a trade-off between the potential benefit from insuring herself against losses and the cost of doing so: selling a large share of her portfolio and giving up the equity premium. As the stock price rises further, and surplus wealth keeps increasing, the cost of switch to the portfolio insurance rule becomes smaller: the investor doesn’t have to sell as many stocks. Therefore, as the price rises enough she eventually switches. This generates a behavior

consistent with the disposition effect: investors have a larger tendency to sell their winners and to hold on to their losers (see Shefrin and Statman (1985) and Odean (1998), among others). In addition, this provides a rational motivation for portfolio insurance strategies, and identifies the conditions under which investors are more or less likely to follow these strategies. Finally, loss-averse investors will abstain from holding equities unless they expect the equity premium to be quite large, and therefore these preferences can help to explain the low stock market participation rates observed in the data.

Heterogeneity in surplus wealth across investors generates trading volume, even in a perfect information setting. We consider a general equilibrium model with two types of investors: one type with power utility and another type exhibiting loss-aversion. This corresponds to a symmetric information version of the model in He and Wang (1995), but with CRRA and Loss-Averse investors instead of CARA investors. Alternatively, it also corresponds to a discrete-time version of the models in Grossman and Zhou (1996) or Basak (1995), but where the demand for portfolio insurance is endogenous and time varying. Basak (2002) also develops a model in which the demand for portfolio insurance is endogenously generated by the investor's preferences. Our models differ because we consider a different preference specification, and because he studies the implications for stock return volatility and for risk premia, while we are concerned with trading volume and with characterizing the portfolio rules specifically generated by the loss-aversion preferences.

The equilibrium model is solved numerically, and yields two main results. First, loss-averse investors can generate a significant degree of trading volume even if they have homogeneous preferences and even if they are a small fraction of the population of investors. Second, when the loss-averse investors are following the generalized portfolio insurance strategy, then trading volume is positively correlated with stock return volatility. Intuitively, when the demand for portfolio insurance is stronger, the aggregate demand for stocks becomes more elastic therefore increasing both the volatility of returns and trading volume. This is the same mechanism as in Grossman and Zhou (1996), and it is consistent with the empirical evidence (see Andersen (1996), Jones, Kaul and Lipson (1994) or Gallant, Rossi and Tauchen (1992)). However, in our model this relationship is not always present since neither is the demand for portfolio insurance. When loss-averse investors are switching strategies,

either from or to the portfolio insurance rule, the relationship between volume and volatility reverses. Consider the case in which the investor is switching to the portfolio insurance rule. The optimal amount of trading is now a negative function of her surplus wealth, since the higher the level of surplus wealth the smaller the amount of stocks that she is required to sell to obtain insurance. The same logic applies in the reverse case, when the investor is switching away from the portfolio insurance rule. This suggests a non-linear relation between the two variables: volume and volatility.²

Recent economic literature has studied some of the implications of loss aversion (see Shiller (1998) or Shleifer (1999) for detailed surveys). Benartzi and Thaler (1995) provide an explanation for the portfolio allocation puzzle (the flip-side of the equity premium puzzle) assuming that investors are loss averse and that they only evaluate their portfolios infrequently, a combination defined as myopic loss aversion. Shumway (1997) uses the same set-up to explain the cross-sectional distribution of expected returns. Epstein and Zin (1990) introduce first-order risk aversion in the consumption CAPM model³ while Lien (2001) studies the implications of loss aversion for futures hedging. Finally, Berkelaar and Kouwenberg (2001) derive closed form solutions for the optimal portfolio choice of a loss-averse investor, assuming a complete markets setting, while Barberis, Huang and Santos (2001) extend the consumption CAPM by assuming that investors derive utility not only from consumption but also from changes in the value of their risky asset holdings. A combination of loss aversion and influence from prior outcomes (as suggested by the evidence from Thaler and Johnson (1990)) determines the preferences over this second component. We differ from these models by considering a set-up with heterogeneity and studying the trading volume implications.⁴ Additionally we consider a pure loss aversion model, in which investors are risk averse in the

²Wang (1993, 1994) and He and Wang (1995) generate positive contemporaneous correlation between stock return volatility and stock turnover with models of differential and asymmetric information. The same result is obtained by Shalen (1993) in a model with dispersion of beliefs, by Brock and LeBaron (1996) in a model with learning, and by Grossman and Zhou (1996) in a model with (exogenous) portfolio insurance.

³More recent models with first-order risk aversion include Bekaert, Hodrick and Marshall (1997), Ang, Bekaert and Liu (2000) and Lien and Wang (2003).

⁴Berkelaar and Kouwenberg (2001) do not study the equilibrium implications of their model, while Barberis, Huang and Santos (2001) use a representative agent set-up.

domain of gains and risk loving in the domain of losses. In the model of Barberis, Huang and Santos (2001), following prior losses, investors actually become even more risk averse. As will show, one consequence of this distinction is that the model in this paper rationalizes a behavior consistent with the disposition effect, while the model in Barberis, Huang and Santos (2001) generates the opposite pattern.

Section 2 describes loss aversion, derives the portfolio allocation behavior of a loss-averse investor, and establishes some partial equilibrium results. Section 3 studies the equilibrium implications of a model with these investors, with a special focus on trading volume. Section 4 concludes and suggests some future work.

2 Optimal portfolio choice with loss aversion

This section studies the optimal portfolio allocation of an investor that exhibits loss aversion. The benchmark used for comparison purposes will be the CRRA case, standard in the intertemporal portfolio choice literature.

2.1 Characteristics of loss aversion

Loss aversion is defined by three properties. First, wealth is measured relative to a given reference point. Second, the decrease in utility implied by a marginal loss (relative to the reference point) is always larger (in absolute value) than the increase in utility resulting from a marginal gain.⁵ Third, although agents are risk averse in the domain of gains, they are risk loving in the domain of losses. A typical utility function would be

$$V^0 \equiv \begin{cases} V_G \equiv \frac{(W-\Gamma)^{1-\gamma}}{1-\gamma} & W \geq \Gamma \\ \lambda V_L \equiv -\lambda \frac{(\Gamma-W)^{1-\gamma}}{1-\gamma} & W < \Gamma \end{cases} \quad (1)$$

where Γ denotes the reference point of the investor, and λ is a positive number greater than 1, that determines the degree of first-order risk aversion.

⁵This property is defined as first-order risk aversion (Epstein and Zin (1990)) and differs from “normal” risk aversion because it holds for infinitesimal gains and losses.

A limitation of this specification is that it implies that marginal utility is decreasing as wealth approaches zero. In an extended framework that avoids this problem the utility function is given by

$$V \equiv \begin{cases} V_G \equiv \frac{(W-\Gamma)^{1-\gamma}}{1-\gamma} & W \geq \Gamma \\ \lambda V_L \equiv -\lambda \frac{(\Gamma-W)^{1-\gamma}}{1-\gamma} & \underline{W} < W < \Gamma \\ V_{BL} \equiv \frac{W^{1-\rho}}{1-\rho} - \left(\lambda \frac{(\Gamma-\underline{W})^{1-\gamma}}{1-\gamma} + \frac{\underline{W}^{1-\rho}}{1-\rho} \right) & W \leq \underline{W} \end{cases} \quad (2)$$

where $\underline{W}$ identifies the level of wealth beyond which the utility function becomes concave (again).^{6,7} This extended set-up allows for the fact that, for big enough losses ($W < \underline{W}$), decreasing marginal utility (of consumption) eventually dominates the psychological effect of the loss. This puts a limit on the amount of risk that the investor is willing to take, whenever she is in a losing position. This is the specification used in our analysis, and it is shown in Figure 1. The vertical axis crosses the horizontal axis at the level of wealth that corresponds to the reference point.⁸

In this specification V is continuous and everywhere differentiable except at $\underline{W}$ and at the reference point (Γ). The non-differentiability at the reference point is a crucial property of loss aversion, while the non-differentiability at $\underline{W}$ is a feature of the specification considered here and it will only affect the technical conditions for some of the results. Marginal utility is always positive but it is increasing in the range of moderate size losses ($[\underline{W}, \Gamma]$). Consistent with Tversky and Kahneman (1992), we consider $\gamma \in [0, 1]$, implying that $V(\Gamma, \Gamma) = 0$. The level of wealth $\underline{W}$ can't be calibrated from the available empirical evidence but for most of this paper that will not be required.

⁶For levels of wealth below $\underline{W}$ the properties of the utility function are given by $\frac{W^{1-\rho}}{1-\rho}$. The two extra terms are just a constant, required to make V a continuous function at $\underline{W}$.

⁷ $\underline{W}$ will be modeled as constant. Alternatively we could specify it as a function wealth (which we will do with the reference point), but this would only add to the algebra and notation in the paper, without changing its results.

⁸Since $\underline{W}$ is a constant, the value function depends on two variables: W, Γ . Therefore, when we wish to specify its arguments, we will write $V(W, \Gamma)$. Likewise we can also write $V_G(W, \Gamma)$ or $V_L(W, \Gamma)$. However, since W and Γ enter linearly in V_G and V_L , we will often use the notation $V_G(W - \Gamma)$ or $V_L(W - \Gamma)$ as it is typically more revealing.

2.2 Optimal portfolio allocation

The analysis in this section considers a static portfolio choice problem, that provides intuition for the results that follow.

In date one the investor chooses how to allocate a given financial wealth, between two assets: one risky and the other one riskless. In date two she liquidates her investment and derives utility from her terminal wealth.

The full static problem (SP) is specified by

$$\begin{aligned} & \max_{\alpha_1} EV(W_2, \Gamma_2) \\ & \text{s.t.} \\ & W_2 = (R_2\alpha_1 + (1 - \alpha_1)R_f)W_1 \end{aligned} \tag{3}$$

where α is the share of wealth invested in the risky asset, R_2 is the return on the risky asset, and R_f is the return on the safe asset.

For simplicity we start by considering a binomial model:⁹

$$R_2 = \begin{cases} R^+ & \text{with probability 0.5} \\ R^- & \text{with probability 0.5} \end{cases} \tag{4}$$

with $R^+ > R_f > R^-$ and

$$R^+ + R^- > 2R_f \tag{5}$$

so that the expected excess return on the risky asset is positive.¹⁰

The dynamics of reference point (Γ) are given by:

$$\Gamma_t = (1 - \theta)R_f\Gamma_{t-1} + \theta W_t \tag{6}$$

⁹In the two-state case it is possible to characterize the solution analytically, and in certain cases derive it in closed form, while otherwise it must be obtained numerically. The results obtained for the two-state case are also valid for more general versions of the problem, namely

$$\ln(R_2) \sim N(\mu, \sigma_R^2) .$$

¹⁰The notation R^i will be used to define the risky asset's return in state i ($R^i = R^+$ or $R^i = R^-$).

with $\theta \in [0, 1)$, so that the reference point is a non-decreasing function of the investor's current wealth. The parameter θ determines the speed of adjustment.¹¹ The reference point is adjusted by the risk-free rate because, even if the stock price remains unchanged for a given period of time, it is plausible that the investor will start considering this as a loss since she could have earned a riskless return instead. The notations $\Gamma_t(R^i)$ and W_t^i will be used to define, respectively, the value of the reference point and the wealth level in state i (when the return on the risky asset is R^i).

2.2.1 Portfolio allocation with CRRA utility

It will be instructive to compare the results for the loss aversion case with the ones obtained for the CRRA case. The CRRA preferences are given by

$$U \equiv \frac{W^{1-\gamma}}{1-\gamma} . \quad (7)$$

The first proposition characterizes the solution of problem (SP) when the investor has CRRA utility. Changes in the current stock price (P_1) lead to changes in the returns in each state of nature. For the purpose of this section those effects are not important. Therefore, we will study price change accompanied by changes in the expectations of future dividends, such that the distribution of the return process remains unchanged.¹²

For a given portfolio composition with current positive stock holdings, a change in P_1 implies a change in current wealth. In the case of CRRA utility this does not affect the optimal portfolio allocation so, holding expectations of future returns constant, the share invested in stocks is independent of the current stock price.

Proposition 1 *If the investor's preferences are given by (7) then*

i) the optimal portfolio allocation (α_1^) in problem P is independent of W and given by:*

$$\alpha_1^* = \frac{R_f(K-1)}{R^- - R_f - K(R^+ - R_f)} \quad (8)$$

¹¹Note that we must have $\theta < 1$ since for $\theta = 1$ we always have $\Gamma_t = W_t$ and therefore, from equation (2) we get $V \equiv 0$.

¹²The results in this section, namely the shapes of the demand curves, have also been derived assuming that only the current stock price changes, while everything else (including expectations about the future) remains unchanged. This, however, only adds to the algebra thus making the crucial effects less clear.

where

$$K = \left(\frac{R_f - R^-}{R^+ - R_f} \right)^{1/\gamma} . \quad (9)$$

ii) Let P_1 denote the price of the risky asset in period 1, then

$$\frac{\partial \alpha}{\partial P_1} \big|_{R^+, R^-} = 0 . \quad (10)$$

Proof: see appendix A.

Since we have a positive expected equity premium (from equation (5)), then $K < 1$ and $\alpha_1 > 0$. As R^- converges to R_f we have $K \rightarrow 0$ and therefore $\alpha_1 \rightarrow \infty$.

It is important to clarify the distinction between the demand curve studied here and the one considered in the empirical literature on the slope of demand curve for stocks (see Shleifer (1986)). This literature looks at market demand curves for individual stocks and is concerned with the degree of substitutability between alternative assets, namely how stock prices react when the relative supply of these assets changes (even if no new information is released). The demand curve implicit in proposition 1 (and others to follow) is an *individual* demand curve for risky assets as a whole, and the investor's wealth is being changed as the stock price changes.

2.2.2 Portfolio allocation with loss aversion and zero surplus wealth

The following proposition characterizes the portfolio allocation rule of a loss-averse investor with zero surplus wealth. In particular this is the situation of an investor that is out of the market and is currently contemplating whether to invest some portion of her wealth in stocks.

When initial surplus wealth is zero the investor will only hold stocks if the financial gain obtained in the good state is sufficiently larger than the financial loss obtained in the bad state, since the marginal utility for losses exceeds the marginal utility for gains. Equation (11) gives a necessary and sufficient condition for this to be true. This participation constraint is more binding as the equity premium decreases, or as the degree of loss aversion increases. Since marginal utility decreases with the size of the loss, if the investor is willing to accept a

small loss then she will also be willing to accept a big one. This logic is valid until the loss is sufficiently large and $W^- < \underline{W}$, when eventually an optimum is reached.

Proposition 2 *Assume that the investor's preferences are given by V , with $W_1 = \Gamma_1$. Then the global optimum for problem SP , $\alpha_1^*(W_1, \Gamma_1, P_1, \underline{W})$ is equal to 0 unless*

$$R^+ - R_f > \lambda(R_f - R^-) \quad (11)$$

in which case $\alpha_1^(W_1, \Gamma_1, P_1, \underline{W})$ is implicitly defined by*

$$[W^+ - \Gamma(R^+)]^{-\gamma}(R^+ - R_f) = (W^-)^{-\rho}(R_f - R^-) . \quad (12)$$

Proof: see appendix A.

The result in proposition 2 suggests that, even if the expected equity premium is positive, investors might not be willing to hold stocks, depending on the specific parameters of the utility function. Since $\lambda > 1$ condition (11) defines a strictly positive lower bound on the expected equity premium. This is a direct implication of first-order risk aversion, and it can help to explain why the majority of households in the population do not invest in equities. In fact, if we take $R_f = 2\%$, and assume a binomial model for stock prices with expected return equal to 8% and standard deviation equal to 15%, then condition (11) is only satisfied if we assume $\lambda < 2.25$ which is exactly the value suggested by the experimental evidence from Tversky and Kahneman (1992). In other words, with $\lambda = 2.25$, this model implies that households should not invest in equities.

2.2.3 Portfolio allocation with loss aversion and negative surplus wealth

The next proposition still considers a loss-averse investor, but studies the case in which surplus wealth is non-positive, although still above $\underline{W}$. Condition (13) imposes a lower bound on W_1 to rule out cases in which the initial wealth is very close to $\underline{W}$.¹³

¹³This condition guarantees that, for any W_1 above this lower bound, if $[\alpha(R^- - R_f) + R_f]W_1 = \underline{W}$ then $[\alpha(R^+ - R_f) + R_f]W_1 > \Gamma_1$, i.e., if wealth in the bad state equals $\underline{W}$ then wealth in the good state exceeds the reference point.

Proposition 3 Assume that the investor's preferences are given by V , with $\widetilde{W} < W_1 \leq \Gamma_1$ ¹⁴, where $\widetilde{W}$ is defined by

$$\widetilde{W} = \frac{1}{1+\phi}\Gamma_1 + \frac{\phi}{1+\phi}\underline{W} \quad (13)$$

with

$$\phi = \frac{R^+ - R_f}{R_f - R^-} \quad (14)$$

then there exists a global optimum for problem SP , $\alpha_1^*(W_1, \Gamma_1, P_1, \underline{W})$ implicitly defined by equation (12).

Proof: see appendix A.

The optimal portfolio rule is the one identified in proposition 2. Once the investor is in a losing position she is risk-loving and therefore she will always be willing to invest in stocks (since they have a higher expected return and higher risk than the safe asset). For levels of wealth below $\underline{W}$ marginal utility is again decreasing and this eventually imposes a limit on the amount of risk that the investor is willing to take. The sign of $\frac{\partial \alpha_1^*}{\partial P_1} \big|_{R^+, R^-}$ is in general undetermined as it will become clear below.

2.2.4 Portfolio allocation with loss aversion and positive surplus wealth

The following propositions characterize the optimal portfolio share invested in stocks when the investor exhibits loss aversion and surplus wealth is strictly positive. Proposition 4 identifies and characterizes a local optimum for this problem. This solution fully protects the investor from losses, as she keeps her portfolio allocation to stocks sufficiently low such that, even in the worst state of nature, her wealth is still above the reference point. The conditions under which this strategy is also a global optimum are discussed in proposition 5.

Proposition 4 If the investor's preferences are given by V and $W_1 > \Gamma_1$ then there exists a (local) optimum for problem SP : $\alpha_1^*(W_1, \Gamma_1, P_1)$ such that

i)

$$\alpha_1^* = \frac{W_1 - \Gamma_1}{W_1} \frac{R_f(K-1)}{R^- - R_f - K(R^+ - R_f)} \quad (15)$$

¹⁴Condition (13) below implies that $\widetilde{W} > \underline{W}$.

where, as before, K is given by (9).

ii) Let P_1 denote the price of the risky asset in period 1, then

$$\frac{\partial \alpha_1^*}{\partial P_1} \big|_{R^+, R^-} > 0$$

with $\frac{\partial \alpha_1^*}{\partial P_1} \big|_{R^+, R^-} \rightarrow 0$ as $\theta \rightarrow 1$.

Proof: see appendix A.

This expression reduces to the one obtained in the CRRA case when the reference point is equal to zero, and converges to it as surplus wealth rises. Changes in the current stock price affect the investor's optimal portfolio allocation, even for given expectations of future returns, because they change the investor's surplus wealth, and therefore they change her risk aversion. As wealth increases (for a given reference point) the investor becomes less risk averse and therefore she increases her risk exposure. Conversely, as the reference point rises, for a given level of wealth, risk aversion increases and the optimal portfolio allocation to stocks is reduced. As surplus wealth falls towards zero, the optimal allocation to stocks also converges to zero, as the investor becomes locally (almost) infinitely risk-averse.

Naturally the magnitude of $\frac{\partial \alpha_1^*}{\partial P_1}$ is going to depend on θ . If as the stock price increases the reference point remains constant, then surplus wealth changes only because current wealth has changed, and therefore this leads to a higher α : the portfolio share invested in the risky asset is a positive function of the stock price. Consider now the limit case in which $\theta \rightarrow 1$ and therefore $\frac{\partial \Gamma_t}{\partial P_t} \rightarrow \frac{\partial W_t}{\partial P_t}$. In this case an increase in the stock price leaves surplus wealth unchanged and therefore α_1 is independent of P_1 , as in the CRRA case. In general, the larger is θ , the smaller $\frac{\partial \alpha_1^*}{\partial P_1} \big|_{R^+, R^-}$ is.

The solution discussed above is consistent with a portfolio insurance strategy: the investor tries to prevent current wealth from falling below the reference point. Benninga and Blume (1985) find that in a complete markets setting “the end-of-period utility function of an investor who insured his or her portfolio at some level would have to exhibit an unbounded coefficient of relative risk aversion below the insurance level and decreasing relative risk aversion above that level.” These conditions are almost perfectly satisfied by a loss-averse investor with positive surplus wealth. Her marginal utility converges to infinity as surplus

wealth falls towards zero. However, marginal utility also decreases rapidly as wealth falls below the reference point, and this suggests that a portfolio insurance type strategy might not be globally optimal. This issue will be re-considered later when studying the (global) optimality of solution identified in proposition 4. In what follows the solution from proposition 4 will be referred to as a generalized portfolio insurance (GPI) rule (for the reasons discussed above and following the terminology of Leland (1980)).

The next proposition presents a necessary and sufficient condition under which the optimum from proposition 4 is a global optimum, and identifies the correct global solution for the case in which this condition fails. The intuition behind this result is the following. Proposition 4 identifies the optimal portfolio allocation for an investor with positive surplus wealth and subject to the constraint that her terminal wealth always exceeds the reference point. This portfolio allocation was shown to be a local optimum for problem SP . However if the investor is willing to tolerate a positive probability of a loss then the alternative candidate for an optimum is given by proposition 5, since the same reasoning that was discussed then still applies: if the investor is willing to accept a small loss then she will also be willing to accept a larger one. Condition (16) below merely compares the value of the two alternative optima to determine which one is the global solution.

Proposition 5 *Consider that the investor's preferences are given by V and $W_1 > \Gamma_1$, and consider the following condition*

$$EV[\bar{\alpha}(R_2 - R_f) + R_f]W_1 - \Gamma(R_2) < EV[\tilde{\alpha}(R - R_f) + R_f]W_1, \Gamma(R) \quad (16)$$

where $\bar{\alpha}$ is the optimum defined in proposition 2 by equation (12), and $\tilde{\alpha}$ is the optimum defined in proposition 4 by equation (15). Then

i) the optimal portfolio rule for problem SP is given by

$$\alpha^* = \begin{cases} \tilde{\alpha} & \text{if condition (16) holds} \\ \bar{\alpha} & \text{if condition (16) does not hold} \end{cases}, \quad (17)$$

ii) a higher (smaller) W_1 makes condition (16) more (less) likely to hold.

Proof: see appendix A.

Part ii) of this proposition states that as the individual's surplus wealth rises, she is more likely to choose the optimum from proposition 4. So, for low levels of surplus wealth, we expect the investor not to follow the generalized portfolio insurance rule, just like she must do when she first invests in stocks. The reason for this behavior is simple: this rule has a very high cost when surplus wealth is small. However, as surplus wealth increases and such cost is reduced, the investor will eventually switch so as to guarantee herself positive surplus wealth, in all states of nature.

Naturally, the “cutoff” between the two strategies will depend on both the investor's preferences and the expected equity premium. So it is possible to calibrate it, either by changing the parameters for the return process or by varying $\underline{W}$.

2.2.5 Demand curve for stocks for a loss-averse investor

Figure 2 plots the demand curve for the loss-averse investor, for different values of θ . As before, the vertical axis crosses the horizontal axis at the price level for which surplus wealth equals zero.¹⁵

Consider first the price range to the RHS of the discontinuity point. Over this range the investor's surplus wealth is non-negative and, as shown in proposition 4, she follows what was defined above as a generalized portfolio insurance (GPI) rule. However, as shown in proposition 5, by avoiding losses the investor must pay a price: the foregone expected return, implied by the lower portfolio share allocated to the risky asset. Eventually, if this cost is very high, she will choose to accept a positive probability of a loss so as to be able to benefit from the high expected return. The cost becomes quite high as the investor's surplus wealth converges to zero since the GPI strategy implies that the wealth share invested in stocks should also go to zero in this case. This generates the discontinuity in the demand

¹⁵The formulation of problem (SP) assumes that the distribution for R_2 has a continuous support. For tractability all the propositions were stated and proved for the simpler binomial model. These results are also valid in the more general case, as can be seen numerically. This can be done by approximating distribution for $Ln(R_2)$ using Gaussian quadrature and obtaining the portfolio rule using a standard grid search algorithm to deal with the non-convexity of the objective function.

function as the investor eventually switches strategies. It is important to note that, in the domain of gains, the loss-averse investor exhibits decreasing relative risk aversion, and therefore this demand curve is not inconsistent with the observation that wealthier individuals invest a larger fraction of their wealth in risky assets.

As shown in proposition 4, an increase in θ reduces the slope of the demand for stocks under the GPI strategy. However, to the left of the discontinuity point, changing θ actually affects the level of the demand curve, and not just the slope. Even when surplus wealth is zero the two demand curves don't coincide. This occurs because the level of surplus wealth obtained in each state is going to depend on the speed at which the reference point adjusts. For a higher θ , a given (positive) α generates both a smaller gain and a smaller loss, respectively in the good and bad states. Therefore to keep the same risk exposure, the share invested in stocks must rise.

2.2.6 The disposition effect

This demand function is consistent with the disposition effect: investors tend to sell winners and hold on to losers (Shefrin and Statman (1985)).¹⁶ However, the evidence in favor of the disposition effect is at the individual stock level, while the predictions of this model hold for risky asset holdings as a whole. Deriving the disposition effect in this framework requires one additional assumption: mental accounting for the holdings of each individual stock.

The results presented here also have important implications for the identification of the determinants and of the evolution of reference points over time. Odean (1998) and Heath, Huddart and Lang (1999) try to identify the implied reference points of the investors by determining the cut-off price, beyond which the probability of selling jumps. According to our results these estimates of reference points are biased upwards as the discontinuity in the probability of selling occurs when surplus wealth is already positive and not when it is zero. Investors can't optimally sell their stocks when surplus wealth becomes marginally positive, because then they would have had no incentive to buy them in the first place.

¹⁶See Odean (1998), Heath, Huddart and Lang (1999), Locke and Mann (1999), Grinblatt and Keloharju (2000, 2001), or Rangelova (2001) for recent empirical evidence on the disposition effect.

3 Equilibrium and trading volume

This section considers a two-period equilibrium version of the previous problem, and studies the implied trading volume patterns. Solving a rational expectations equilibrium with heterogeneous investors is quite problematic since the expectations of future prices depend on the distribution of all the relevant state variables which in our case are the stock holdings, levels of wealth and surplus wealth for all investors, in every possible state of nature in the future. These naturally depend on the current decisions/allocation which in turn depend on these expectations. This section deals with this problem by considering a two-period binomial model so that it becomes feasible to solve it numerically. Appendix B discusses a model with T -periods which requires additional assumptions: myopic portfolio behavior (investors choose their portfolio allocations assuming that these are buy-and-hold strategies) and no adjustment of the reference points. In this case the investors don't care about the distribution of future prices (except for the distribution of the terminal price which is exogenous) and the problem can also be solved numerically. The main results of the two models are the same.

The structure is derived from He and Wang (1995), but in a symmetric information context. Since in this model there is a group of investors with a time varying demand for portfolio insurance it can also be linked to Grossman and Zhou (1996) or Basak (1995).¹⁷ However, in the loss aversion model, the demand for portfolio insurance is not constant, becoming a function of the relevant state variables.

3.1 Set-up

There are two types of investors: loss-averse investors, and CRRA investors. The loss-averse investors solve the following problem (DP):

$$\max_{\{\alpha_0, \alpha_1\}} E_0 V(W_2, \Gamma_2) \quad (18)$$

¹⁷The model in Basak (1995) allows for intermediate consumption unlike the one in Grossman and Zhou (1996) or the one in this paper.

s.t.

$$\Gamma_t = (1 - \theta)R_f\Gamma_{t-1} + \theta W_t \quad (19)$$

$$W_{t+1} = P_{t+1}S_t + R_f B_t \quad t = 0, 1 \quad (20)$$

$$\alpha_t \in [0, 1] \quad t = 0, 1 \quad (21)$$

$$P_2 = \omega_2 + \omega_1 \quad (22)$$

$$\omega_t = \begin{cases} \omega^H & \text{with probability 0.5} \\ \omega^L & \text{with probability 0.5} \end{cases} \quad t = 1, 2 \quad (23)$$

with $\omega^H > \omega^L$, and where S_t and B_t denote, respectively, risky asset holdings and riskless asset holdings at time t . The short-selling constraint on the portfolio allocation (equation (21)) limits the amount of risk-taking in the domain of losses and is motivated by the desire to make the results independent of the choice of $\underline{W}$, which can't be calibrated from the data.

The CRRA investors solve the same problem but with $U(W_2)$ replacing $V(W_2, \Gamma_2)$. The policy rules for the second period have already been derived, in proposition 1 for the CRRA investors, and in propositions 2 through 5 for the loss-averse investors. The first-period policy rules are obtained numerically as the problem is solved backwards.¹⁸

The market clearing condition is

$$\alpha_t^{LA} W_t^{LA} + \alpha_t^C W_t^C = P_t \bar{S} \quad t = 1, 2 \quad (24)$$

where $\bar{S}$ represents the total (exogenous) supply of stocks, P_t is the stock price at time t , and the superscripts LA and C are now used to identify, respectively, the loss-averse investors and the CRRA investors. As in He and Wang (1995) the supply of the riskless asset is perfectly elastic at the given risk-free rate.

Note that, if stocks are to have a positive expected return as of date 0, then it must be the case that

$$\frac{\omega^H + \omega^L}{P_0} > (R_f)^2 \quad (25)$$

and additionally, so that stocks don't fully dominate bonds at date 0, we must also have

$$\frac{\omega^L + \omega^L}{P_0} < (R_f)^2 \quad (26)$$

¹⁸The details on the numerical solution are given in appendix B.

3.2 Equilibrium

In this subsection we discuss the implications of this model for trading volume. There are two main results. First, the presence of the loss-averse investors can generate a significant degree of trading volume even if they have homogeneous preferences and even if they are a small fraction of the population of investors. Second, when the loss-averse investors are following the GPI strategy then trading volume is positively correlated with stock return volatility. However, when they are switching between strategies, the sign of this correlation reverses. This suggests a non-linear relation between the two variables.

3.2.1 Loss-averse investors with low initial surplus wealth

This is the case in which initial surplus wealth for the loss-averse investor is negative, zero or only marginally positive, such that her optimal portfolio rule is NOT given by the GPI strategy. As shown in figure 3a, the short-selling constraint is initially binding, and she only holds stocks. If $\omega^1 = \omega^L$ then the stock price falls, the loss-averse investor remains fully invested in stocks, and there is no trading volume. If $\omega_1 = \omega^H$ then the stock price rises and, given our calibration, the loss-averse investor switches to the GPI strategy.¹⁹

The benchmark results are shown for the following preference parameters: $\lambda = 1.5$, $\gamma = 0.5$ for the loss-averse investors and $\gamma = 5$ for the CRRA investors, and $\theta = 0.5$. Results for different values of θ are presented below. We considered values of λ ranging from 1.25 to 1.75. As was shown before, even small values of λ require large risk premia to keep the loss-averse investors in the market. As for γ we considered values going from 0.2 to 0.8. In all cases the results are found to be robust.

Figure 3b plots the stock return in the high state as a function of ω^H , for different values of the ratio W_0^C/W_0^{LA} . As argued above, the model was calibrated so that the returns in this state are high enough to induce the loss-averse investors to switch to the GPI strategy. As expected a higher value of ω^H is associated with a higher return since its impact on P_1^H

¹⁹Considering values of ω^H such that this switch doesn't occur would be of very limited interest since the model wouldn't generate any trading volume.

is higher than its impact on P_0 . As we increase W_0^C the stock return falls. This occurs because in this region the aggregate demand curve is actually positively sloped. Once the loss-averse investors are following the GPI strategy, a higher stock price will actually increase their demand for stocks since it increases their surplus wealth and therefore reduces their risk aversion. As the loss-averse investors become more negligible the stock price doesn't have to increase so much to generate sufficient demand. In the limit we obtain the volatility of a model with only CRRA investors.

Figure 3c plots trading volume as a function of ω^H and for different values of the ratio W_0^C/W_0^{LA} . In all cases we find that trading volume is negatively correlated with ω^H . Since, from figure 3b, we know that a higher value of ω^H corresponds to a higher return in the good state, and since changing ω^H has a much smaller impact on P_1^L than on P_1^H , then trading volume is negatively correlated with stock return volatility. Remember that trading volume occurs because the loss averse investors are partially liquidating their positions, and switching to the GPI strategy. Consequently, the larger the price change, the smaller the amount of stocks that these investors have to liquidate.

A less intuitive result is that, as we increase W_0^C trading volume also increases. A higher W_0^C generates two effects. On the one hand the loss-averse investors hold less shares and therefore should generate less trading volume. But, as we saw in figure 3b, this also reduces the stock return and therefore reduces surplus wealth and the optimal stock holdings of the loss-averse investors. Naturally as the share of loss-averse investors becomes very small the first effect should dominate and therefore we should observe a non-linear relationship between W_0^C and trading volume. However, as we increase the initial weight of the CRRA investors the risk premium in the economy falls, and the loss-averse investors are eventually excluded from the market. In our model this occurs before the reversal actually takes place. In other words, the reversal occurs in a discontinuous way. This last result (the discontinuity) is a specific feature of our model, as we have no heterogeneity between loss-averse investors and therefore all trades must take place with the CRRA investors. However the main result is particularly interesting as it suggests that an economy with loss-averse investors will be able to generate trading volume, even if these investors are not a large fraction of the relevant population.

In figures 3*d* and 3*e* we study the impact of changing the adjustment rate for the reference point (θ). We find that as we increase θ the stock return rises, but not too much. At any given level of wealth, a higher θ implies lower surplus wealth for the loss-averse investor and therefore a smaller level of aggregate demand. Since the demand is positively sloped in this region, we have an increase in the stock price and therefore a higher return. Trading volume also rises with θ , since a higher θ is associated with a lower value of surplus wealth and therefore the loss-averse investor wants to sell a larger share of her stock holdings. The effect of having a higher stock return is clearly second-order since it only occurs exactly to the extent that there is less demand from the loss-averse investors.

3.2.2 Loss-averse investors with high initial surplus wealth

Now we consider the case in which initial surplus wealth for the loss-averse investor is sufficiently positive, such that her optimal portfolio rule in period 0 is given by the GPI strategy. This is shown in figure 4*a*. Under our calibration, if $\omega_1 = \omega^L$ then, as the stock price falls, the loss-averse investor gives up the GPI strategy and invests fully in stocks. Note that, unlike in the previous case, now there is trading volume in this state as well. If $\omega_1 = \omega^H$ then the stock price rises and, given our calibration, the loss-averse investor keeps following the GPI strategy, although her demand for portfolio insurance weakens as surplus wealth has increased.

Figure 4*b* shows trading volume in the high state as function of ω^H (i.e. as a function of the stock return), and for different levels of initial surplus wealth. Now we find that trading volume is positively correlated with stock returns. This occurs because the investor is following the GPI strategy in both periods. Therefore larger returns generate larger fluctuations in surplus wealth and therefore more trading volume. This is consistent with the findings of Grossman and Zhou (1996): when investors are following portfolio insurance strategies, trading volume and stock returns are positively correlated. Figure 8 also shows that the value of initial surplus wealth and trading volume are negatively correlated. A higher value of initial surplus wealth reduces the demand for portfolio insurance and therefore reduces the elasticity of the demand curve.

The level of trading volume in figure 4b is much smaller than the one reported in figure 3c. Trading volume is at its maximum level when the loss-averse investor is switching strategies. Consistent with this we also obtain a very high turnover ratio in the bad state ($\omega^1 = \omega^L$), when the investor is switching away from the GPI strategy. This is shown in Figure 4c, and it is just the reverse of the case considered above. The lower the value of initial surplus wealth the lower the initial stock holdings of the loss-averse investor, and therefore the higher the turnover ratio, as she is now fully invested in stocks. This result is not a consequence of the short-selling constraint. Remember that the portfolio allocation in the domain of losses is roughly independent of the level of surplus wealth, while the GPI rule depends strongly on that level. As before we find that, when the investor is switching strategies, trading volume is negatively correlated with stock returns. Therefore the results suggest that the relation between trading volume and stock return volatility should be non-linear.

4 Conclusion and directions for future work

This paper studies the optimal portfolio allocation behavior of loss-averse investors and its implications for trading volume. The demand function for risky assets is discontinuous and non-monotonic. As surplus wealth reaches a certain threshold, investors sell a large part of their stock holdings and follow a (generalized) portfolio insurance rule, protecting themselves against losses (relative to their reference point). In addition this provides a rational motivation for portfolio insurance strategies, and identifies the conditions under which investors are more or less likely to follow those strategies.

Since the value function exhibits first-order risk aversion, this implies that loss-averse investors will abstain from holding stocks unless they expect the equity premium to be quite high. Simulation results show that this model is able to rationalize the small participation rates observed in the data.

A dynamic model, in the spirit of Grossman and Zhou (1996), Basak (1995, 2000) and He and Wang (1995), (typically) yields positive correlation between stock return volatility and trading volume. When the demand for portfolio insurance increases the aggregate demand for stocks becomes more elastic and at the same time trading volume increases. This generates

the positive correlation between both series. However, in our model this result is not globally valid since the demand for portfolio insurance is not always present. When loss-averse investors are switching strategies the relationship between volume and volatility reverses. This is also the moment in which trading volume is at its peak. Note however that, in a model with heterogeneous loss-averse investors, when some investors are switching strategies it is also when others are more likely to be close to switching as well. But this implies that the demand for portfolio insurance for this second group should be quite high contributing to a higher correlation between volume and volatility. This suggests that we should not expect a discontinuity, as obtained in the model with homogeneous (loss-averse) investors, but rather a smooth transition. In any case a definite prediction for the relationship between trading volume and stock return volatility will be quite hard to make, until such a model is actually solved.

Shiller (1981) and LeRoy and Porter (1981) show that stock prices are too volatile to be explained by realistically calibrated shocks to future cash-flows or moderate changes in discount rates. If investors exhibit loss aversion then moderate changes in wealth can lead to large changes in discount rates which might help explain this puzzle.

Other generalizations would be quite interesting. Allowing for asymmetric information will be very important to develop a more complete model of trading volume and it will also help to determine the quality of market prices as signals of fundamentals. As shown by Grossman and Zhou (1996), models with portfolio insurance have important implications for option pricing. By generating a time-varying demand for portfolio insurance the loss-aversion model should generate some very interesting option pricing dynamics. Deriving these dynamics and testing the model along this additional dimension is another promising direction for future research.

Appendix A: Proofs

Proof of Proposition 1:

i) The first-order condition for this problem is (using the specification for U):

$$0.5[\alpha(R^+ - R_f) + R_f]^{-\gamma}(R^+ - R_f) + 0.5[\alpha(R^- - R_f) + R_f]^{-\gamma}(R^- - R_f) = 0 . \quad (27)$$

Re-arranging this expression we obtain

$$\left[\frac{\alpha(R^+ - R_f) + R_f}{\alpha(R^- - R_f) + R_f} \right]^{-\gamma} = \frac{R_f - R^-}{R^+ - R_f} \Leftrightarrow \quad (28)$$

$$\left[\frac{\alpha(R^- - R_f) + R_f}{\alpha(R^+ - R_f) + R_f} \right] = \left[\frac{R_f - R^-}{R^+ - R_f} \right]^{1/\gamma} . \quad (29)$$

Defining K as

$$K = \left(\frac{R_f - R^-}{R^+ - R_f} \right)^{\frac{1}{\gamma}} \quad (30)$$

and solving for α^* :

$$\alpha^* = \frac{R_f(K - 1)}{R_f(K - 1) + R^- - KR^+} . \quad (31)$$

From the expression (31) we can compute the derivative of α with respect to W_1 , and check that $\frac{\partial \alpha}{\partial W_1} = 0$. ■

ii) By definition we have that

$$\frac{\partial \alpha}{\partial P_1} = \frac{\partial \alpha}{\partial R^+} \frac{\partial R^+}{\partial P_1} + \frac{\partial \alpha}{\partial R^-} \frac{\partial R^-}{\partial P_1} + \frac{\partial \alpha}{\partial W} \frac{\partial W}{\partial P_1} \quad (32)$$

When holding R^+ and R^- constants the first two terms equal zero and we obtain:

$$\frac{\partial \alpha}{\partial P_1} \Big|_{R^+, R^-} = \frac{\partial \alpha}{\partial W} \frac{\partial W}{\partial P_1} \quad (33)$$

and from equation (31) we easily get the result. ■

Proof of Proposition 2:

For $W_1 = \Gamma_1$, the marginal utility from an infinitesimal increase in α (from $\alpha = 0$) is given by:

$$0.5V_G'(0)(R^+ - R_f)W_1 + \lambda 0.5V_L'(0)(R^- - R_f)W_1 \quad (34)$$

where $V_G'(0)$ and $V_L'(0)$ are, respectively, the right-hand-side derivative of V_G at 0, and the left-hand-side derivative of V_L at 0. From their definitions we have that $V_G'(0) = V_L'(0)$,

and therefore, dividing both terms by $0.5W_1$, a necessary and sufficient condition for $\alpha^* > 0$ is that:

$$R^+ - R_f > \lambda(R_f - R^-) . \quad (35)$$

This condition does not depend on α and corresponds to condition (11).

If the function V_L were defined over the whole domain of losses then, under condition (11), there wouldn't be a maximum as the optimal α would tend to infinity. However, for α high enough such that W^- falls below $\underline{W}$, the marginal utility loss is given by $-0.5V_{BL}(W^-)'(R^- - R_f)W_1$. As α increases, $V_G'(W^+ - \Gamma_2(R^+))$ converges to zero, while $V_{BL}(W^-)'(R_f - R^-)$ converges to infinity, so there must exist an α such that²⁰

$$[W^+ - \Gamma_2(R^+)]^{-\gamma}(R^+ - R_f) = (W^-)^{-\rho}(R_f - R^-) \quad (36)$$

and this equation implicitly defines the optimum portfolio allocation. ■

Proof of Proposition 3:

This proof follows closely the proof of proposition 2ii).

Since $\widetilde{W}_1 < W_1 < \Gamma_1$ then the investor's expected utility, for a small α , is given by²¹

$$0.5\lambda V_L(W^+, \Gamma(R^+)) + 0.5\lambda V_L(W^-, \Gamma(R^-)) .$$

Since V_L is a strictly convex function (the agent is risk-loving in the domain of moderate size losses), we know that $\alpha \leq 0$ is not an optimum.

As we consider increasing α , from $\alpha = 0$, the marginal utility gain is given by:

$$0.5\lambda V_L'(W^+, \Gamma(R^+))(R^+ - R_f) + 0.5\lambda V_L'(W^-, \Gamma(R^-))(R^- - R_f) .$$

Again, by the convexity of V_L , we know that this is always positive. When α is large enough

²⁰Note that $V_{BL}'(\underline{W})$ is equal to zero, while $V_G'(W^+ - \Gamma)$ is always strictly positive, and since both are continuous monotonic functions, they must eventually cross.

²¹Note that, by definition, $\widetilde{W}_1 > \underline{W}$:

$$\widetilde{W} = \frac{1}{1+\phi}\Gamma_1 + \frac{\phi}{1+\phi}\underline{W}$$

and we have $\phi > 0$ and $\Gamma_1 > \underline{W}$.

we will have $W^+ \geq \Gamma(R^+)$ and the marginal utility gain is now given by:²²

$$0.5\lambda V'_G(W^+, \Gamma(R^+))(R^+ - R_f) + 0.5\lambda V'_L(W^-, \Gamma(R^-))(R^- - R_f) .$$

And the rest of the proof follows directly from the proof of proposition 2ii). ■

Proof of Proposition 4:

Using the law of motion for the reference point we can write surplus wealth in period 2, $W_2 - \Gamma_2$, as:

$$W_2 - \Gamma_2 = W_2 - (1 - \theta)R_f\Gamma_1 - \theta W_2 = (1 - \theta)[W_2 - R_f\Gamma_1] . \quad (37)$$

And, if we have $W_2 = R_f W_1$ ($\alpha = 0$) this becomes

$$W_2 - \Gamma_2 = (1 - \theta)R_f[W_1 - \Gamma_1] . \quad (38)$$

As a result, if $W_1 > \Gamma_1$ and $\alpha = 0$, then $W_2 > \Gamma_2$, and therefore

$$EV(R_f W_1, \Gamma_2) = V_G(R_f W_1, \Gamma_2) . \quad (39)$$

For α small enough, it will still be true that

$$\alpha_1 R^i W_1 + (1 - \alpha_1)R_f W_1 \geq \Gamma_2, \quad \forall_{R^i \in \{R^-, R^+\}} . \quad (40)$$

Define $\tilde{\alpha}$ from the following condition:

$$\tilde{\alpha} R^- W_1 + (1 - \tilde{\alpha})R_f W_1 = \Gamma_2 . \quad (41)$$

This implies that, for any portfolio allocation in the set $[o, \tilde{\alpha}]$, all payoffs will occur in the domain of gains.

Therefore we know that

$$\max_{\alpha_1 \in [o, \tilde{\alpha}]} EV(\alpha_1 R W_1 + (1 - \alpha_1)R_f W_1, \Gamma_2) \quad (42)$$

is equivalent to

$$\max_{\alpha_1 \in [o, \tilde{\alpha}]} EV_G(\alpha_1 R W_1 + (1 - \alpha_1)R_f W_1, \Gamma_2) . \quad (43)$$

²²From the definition of $\widetilde{W}$, for $W^+ = \Gamma(R^+)$ we still have $W^- > \underline{W}$. In other words, we have assumed that W_1 is “closer to the reference point” than to $\underline{W}$.

Let α^* denote the solution to this problem. Since V_G is a concave function, we know that α^* must satisfy:

$$0.5W_1(R^+ - R_f)(W^+ - \Gamma_2(R^+))^{-\gamma} + 0.5W_1(R^- - R_f)(W^- - \Gamma_2(R^-))^{-\gamma} = 0 \quad (44)$$

with

$$\begin{cases} W^+ = [\alpha^*(R^+ - R_f) + R_f]W_1 \\ W^- = [\alpha^*(R^- - R_f) + R_f]W_1 \end{cases}.$$

Substituting the expressions for $\Gamma_2(R^+)$ and $\Gamma_2(R^-)$ we can re-write the first-order condition as

$$(R^+ - R_f)[(1 - \theta)(W^+ - R_f\Gamma_1)]^{-\gamma} = (R^- - R_f)[(1 - \theta)(W^- - R_f\Gamma_1)]^{-\gamma}. \quad (45)$$

Solving for α^* ,

$$\alpha^* = \frac{(W_1 - \Gamma_1)}{W_1} \frac{R_f(K - 1)}{R^- - R_f - K(R^+ - R_f)} \quad (46)$$

where, as before, K is given by (30).

Finally, since $V'_G(0) = +\infty$, α^* is a local optimum for problem SP . ■

ii) Using the definition of Γ_1 we can re-write equation (46) as:

$$\alpha^* = \frac{((1 - \theta)W_1 - (1 - \theta)R_f\Gamma_0)}{W_1} \frac{R_f(K - 1)}{R^- - R_f - K(R^+ - R_f)}.$$

By definition we know that $\frac{\partial \alpha}{\partial P_1} |_{R^+, R^-} = \frac{\partial \alpha}{\partial W_1}$ so

$$\frac{\partial \alpha}{\partial P_1} |_{R^+, R^-} = \frac{(1 - \theta)R_f\Gamma_0}{(W_1)^2} \frac{R_f(K - 1)}{[R^- - R_f - K(R^+ - R_f)]} > 0 \quad (47)$$

since $\theta \in [0, 1)$. ■

Proof of Proposition 5:

i) Proposition 4 identifies a local optimum for problem SP , for an investor with positive surplus wealth. This optimum is given by equation (46), and let us denote it by $\tilde{\alpha}$.

This was shown to be the optimal solution when we added the constraint $W^- > \Gamma_1$ to problem SP . Now we need to consider whether larger values of α (therefore yielding $W^- < \Gamma_1$) can potentially generate higher utility than $\tilde{\alpha}$.

If the investor is willing to tolerate a positive probability of a loss then, from the proof of proposition 2ii), the alternative candidate for an optimum is given by equation (12). Denote

this alternative optimum by $\bar{\alpha}$. Condition (16) merely compares the utility level given by $\tilde{\alpha}$ and $\bar{\alpha}$ to determine the global optimum. ■

ii) The utility obtained by choosing $\alpha = \tilde{\alpha}$ (the optimum from proposition 4) is :

$$\tilde{V} = 0.5 \frac{(W^-(\tilde{\alpha}) - \Gamma_2(R^-; \tilde{\alpha}))^{1-\gamma}}{1-\gamma} + 0.5 \frac{(W^+(\tilde{\alpha}) - \Gamma_2(R^+; \tilde{\alpha}))^{1-\gamma}}{1-\gamma} \quad (48)$$

while the utility derived from choosing $\bar{\alpha}$ (the optimum defined in condition (12)) is:

$$\bar{V} = -0.5 \frac{(W^-(\bar{\alpha}))^{1-\rho}}{1-\rho} + 0.5 \frac{(W^+(\bar{\alpha}) - \Gamma_2(R^+; \bar{\alpha}))^{1-\gamma}}{1-\gamma} . \quad (49)$$

Using the envelope theorem, and defining $R_P^i(\alpha) = \alpha(R^i - R_f) + R_f$:

$$\begin{aligned} \frac{\partial \tilde{V}}{\partial W_1} = & 0.5(R_P^-(\tilde{\alpha}) - (1-\theta)\theta R_f)(W^-(\tilde{\alpha}) - \Gamma_2(R^-; \tilde{\alpha}))^{-\gamma} + \\ & 0.5(R_P^+(\tilde{\alpha}) - (1-\theta)\theta R_f)(W^+(\tilde{\alpha}) - \Gamma_2(R^+; \tilde{\alpha}))^{-\gamma} \end{aligned} \quad (50)$$

$$\frac{\partial \bar{V}}{\partial W_1} = 0.5R_P^-(\bar{\alpha})(W^-(\bar{\alpha}))^{-\rho} + 0.5(R_P^+(\bar{\alpha}) - (1-\theta)\theta R_f)(W^+(\bar{\alpha}) - \Gamma_2(R^+; \bar{\alpha}))^{-\gamma} . \quad (51)$$

From the first order conditions for $\tilde{\alpha}$ and $\bar{\alpha}$ we know that we can define

$$\tilde{T} \equiv (W^+(\tilde{\alpha}) - \Gamma_2(R^+; \tilde{\alpha}))^{-\gamma}(R^+ - R_f) = (W^-(\tilde{\alpha}) - \Gamma_2(R^-; \tilde{\alpha}))^{-\gamma}(R_f - R^-) \quad (52)$$

$$\bar{T} \equiv (W^+(\bar{\alpha}) - \Gamma_2(R^+; \bar{\alpha}))^{-\gamma}(R^+ - R_f) = (W^-(\bar{\alpha}))^{-\rho}(R_f - R^-) . \quad (53)$$

Using the definitions of $\tilde{T}$ and $\bar{T}$ it is possible to re-write equations (50) and (51) as

$$\frac{\partial \tilde{V}}{\partial W_1} = 0.5\tilde{T} \left[\frac{R_P^-(\tilde{\alpha}) - (1-\theta)\theta R_f}{R_f - R^-} - \frac{R_P^+(\tilde{\alpha}) - (1-\theta)\theta R_f}{R^+ - R_f} \right] \quad (54)$$

$$\frac{\partial \bar{V}}{\partial W_1} = 0.5\bar{T} \left[\frac{R_P^-(\bar{\alpha})}{R_f - R^-} - \frac{R_P^+(\bar{\alpha}) - (1-\theta)\theta R_f}{R^+ - R_f} \right] . \quad (55)$$

Now note that

$$\frac{R_P^-(\alpha)}{R_f - R^-} + \frac{R_P^+(\alpha)}{R^+ - R_f} = -\alpha + \frac{R_f}{R_f - R^-} + \alpha + \frac{R_f}{R^+ - R_f} = \frac{R_f(R^+ - R^-)}{(R_f - R^-)(R^+ - R_f)} \quad (56)$$

which does not depend on α . Also, since $\bar{\alpha} > \tilde{\alpha}$, then

$$(W^+(\tilde{\alpha}) - \Gamma_2(R^+; \tilde{\alpha}))^{-\gamma}(R^+ - R_f) > (W^+(\bar{\alpha}) - \Gamma_2(R^+; \bar{\alpha}))^{-\gamma}(R^+ - R_f) \quad (57)$$

and therefore, combining equations (57), (52) and (53), we have $\tilde{T} > \bar{T}$.

So, defining:

$$H_1 = \frac{R_f(R^+ - R^-)}{(R_f - R^-)(R^+ - R_f)} \quad (58)$$

and

$$H_2 = \frac{(1 - \theta)\theta R_f}{R^+ - R_f} - \frac{(1 - \theta)\theta R_f}{R_f - R^-} \quad (59)$$

we have

$$\frac{\partial \tilde{V}}{\partial W_1} = \tilde{T}(H_1 + H_2) > \bar{T}(H_1 + 0.5H_2) = \frac{\partial \bar{V}}{\partial W_1} \quad (60)$$

since both H_1 and H_2 are positive.

■

Appendix B: Dynamic Model

This is essentially a T -period version of the model in section 3. The structure is derived from He and Wang (1995), but in a symmetric information context.

i) Set-up of the model

As before the supply of the risky asset is exogenously fixed, while the supply of the risk-free asset is perfectly elastic at a given risk-free rate. There are two types of investors: investors with CRRA preferences and loss-averse investors.

The relevant information set/state variable follows a (logarithmic) random walk and the reference point is assumed to be constant during the T periods for reasons discussed below.²³

The full dynamic problem (DTP) is specified by

$$\max_{\{\alpha_i\}_{i=1}^T} EV(W_T, \Gamma_T)$$

s.t.

$$W_{t+1} = P_{t+1}S_t + R_f B_t \text{ for } t = 1, \dots, T \quad (61)$$

$$P_T = \theta_T \quad (62)$$

$$\ln(\theta_{t+1}) = \ln(\theta_t) + \omega_t, \quad \omega_t \sim N(0, \sigma_\omega^2) \quad (63)$$

$$\Gamma_T = R_f^T W_0 \quad (64)$$

$$\alpha_t \in [0, 1], \forall_t \quad (65)$$

where Γ_t is the investor's reference point at time t .

In general this problem can't be solved since the pricing function depends on an infinite number of state variables (the infinite regress problem discussed in Townsend (1983)). This motivates the assumption of a fixed reference point and it also requires one additional assumption: the investor chooses α_t assuming that she can't rebalance her portfolio in the future (myopic portfolio allocation).

²³ T will be set equal to 252, the average number of trading days in a year. Therefore this assumption implies that reference points are constant over one-year periods, just like in the models of Barberis, Haug and Santos (2001) and Benartzi and Thaler (1995).

The other constraint on α_t (equation (65)) limits risk-taking behavior in the domain of losses, therefore making the calibration of $\underline{W}$ and ρ virtually irrelevant. This is particularly helpful since these parameters can't be rigorously calibrated from existing evidence.

The market-clearing dynamics assumed here are the following: in response to excess demand (supply) the market maker will increase (decrease) the stock price.

ii) Numerical Solution

Since the investor solves the problem under the assumption of no rebalancing, the optimal solution exhibits the properties derived in section 2. This is true even in the presence of the short-selling constraint. The distribution for the state variable was approximated using Gaussian quadrature. The state-space was discretized along all the other dimensions, using equally spaced grids with non-binding upper and lower bounds. The distance between any two grid points was determined by an upper bound of 2.5 percentage points for implied change in the optimal portfolio rule (share invested in the risky asset).²⁴

The solution given by equation (15) can be computed directly, while the solution from equation (12) was obtained using a simple fixed-point algorithm. The market clearing condition was solved using a recursive algorithm based on the market clearing rule specified above. In each period the algorithm was started by computing the excess demand at last period's stock price, given the new information set of each group of investors.

After solving the model we simulate 5000 different time series and generate different cross sections from them.²⁵

iii) Data

The predictions of this model are compared against empirical evidence derived from high frequency data, namely daily stock returns and daily trading volume.

The return data correspond to the value-weighted stock return on the NYSE, the AMEX, and the NASDAQ, from July 1962 until December 1996,²⁶ taken from CRSP. The volume

²⁴Increasing the number of grid points (for given upper and lower bounds) did not produce any meaningful change in the results.

²⁵To minimize both the effects of the initial conditions and horizon effects, the cross sections were taken for data points t^* such that $0 < t^* < T$. T was set at 252, the average number of trading days in a year. The results are robust to changes in the value of t^* .

²⁶The volume data starts in July 1962.

data was also taken from CRSP. The measure of volume chosen was the turnover ratio, constructed by aggregating trading volume for individual stocks, measured in dollar units, and dividing it by total market capitalization. Trading volume for each individual stock and its corresponding price were taken from the CRSP daily stock files. Only ordinary common shares and certificates that are traded on the NYSE, the AMEX or the NASDAQ were considered (this implied dropping 7.2% of the original sample). The volume variable reports the number of shares sold on a given day, rounded to the nearest hundred. The price variable corresponds either to the closing price or the average of the bid and ask prices, on that same day (the files also provide information on which one is actually being reported). For roughly 99% of observations in which trading volume is positive, the closing price is the one that is reported. Missing observations can be distinguished from observations with zero trading volume and were dropped from the sample (they correspond to 0.1% of the remaining sample size). Following Campbell, Grossman and Wang (1993), we consider the natural logarithm of the turnover ratio and low frequency patterns were removed by subtracting a one-year backward moving average filter.

iv) Results and empirical evidence

The results with simulated data correspond to the average across 20 different cross sections, each one of them containing 5000 observations. Unlike what one might think, the model does not have many free parameters. Results will be presented for different values for λ and γ and for the percentage of initial wealth of the CRRA investors (ω^C), the only relevant free parameters.

Table A1 reports the correlations between volume and turnover, volume and lagged volume, and turnover and lagged turnover, for both the simulated data and the data CRSP.²⁷ It is not the purpose of this section to match specific moments as this model is too simplified for that. Instead the objective is to show that the model can generate the correct qualitative predictions and that the magnitudes are economically meaningful. For all combinations of parameter values the correlations are strongly positive and significant, consistent with the evidence from CRSP.

²⁷The signs and economic significance of these correlations survive more detailed empirical studies as shown by Gallant, Rossi and Tauchen (1992), Bollerslev, Chou and Kroner (1992) and several others.

Table A1 - Sample correlations from simulated data and from CRSP data

λ	γ	ω^c	Corr(vol,trn)	Corr(trn,Ltrn)	Corr(vol,Lvol)
1.50	0.8	1	0.21	0.44	0.49
1.75	0.8	1	0.21	0.41	0.49
1.50	0.8	2	0.06	0.18	0.44
1.50	0.6	1	0.17	0.62	0.36
CRSP Data:			0.12	0.45	0.19

where: vol - volatility, trn - turnover, Ltrn - lagged turnover, Lvol - lagged volume.
(the standard errors are 0.0109 for the CRSP data and 0.0032 for the simulated data)

When the demand for portfolio insurance is stronger then the aggregate demand for stocks becomes more elastic and trading volume increases. This generates the positive correlation between volume and volatility, just like in Grossman and Zhou (1996). Consistent with the data the loss aversion model also generates persistence in volatility and in trading volume. This persistence occurs because the demand for portfolio insurance rules is itself a persistent process, as it is motivated by the level of surplus wealth.

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Figure 1 - Value function for the loss averse investor

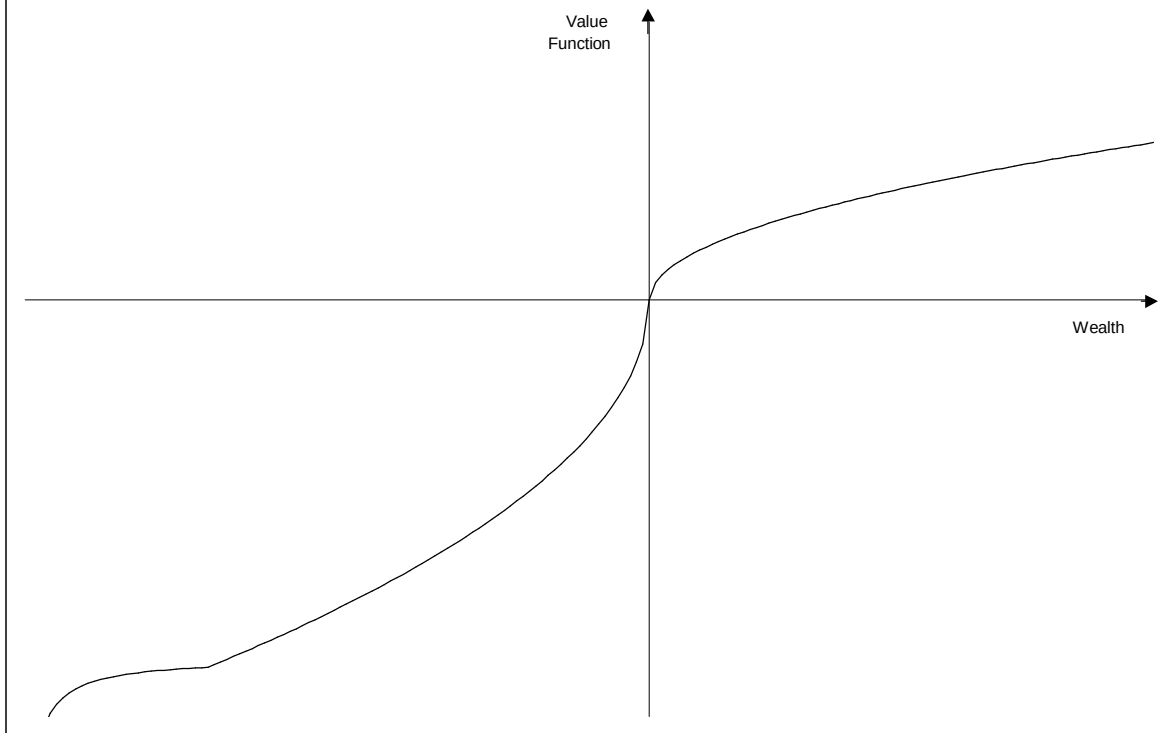


Figure 2 - Demand curve for stocks for the loss-averse investor

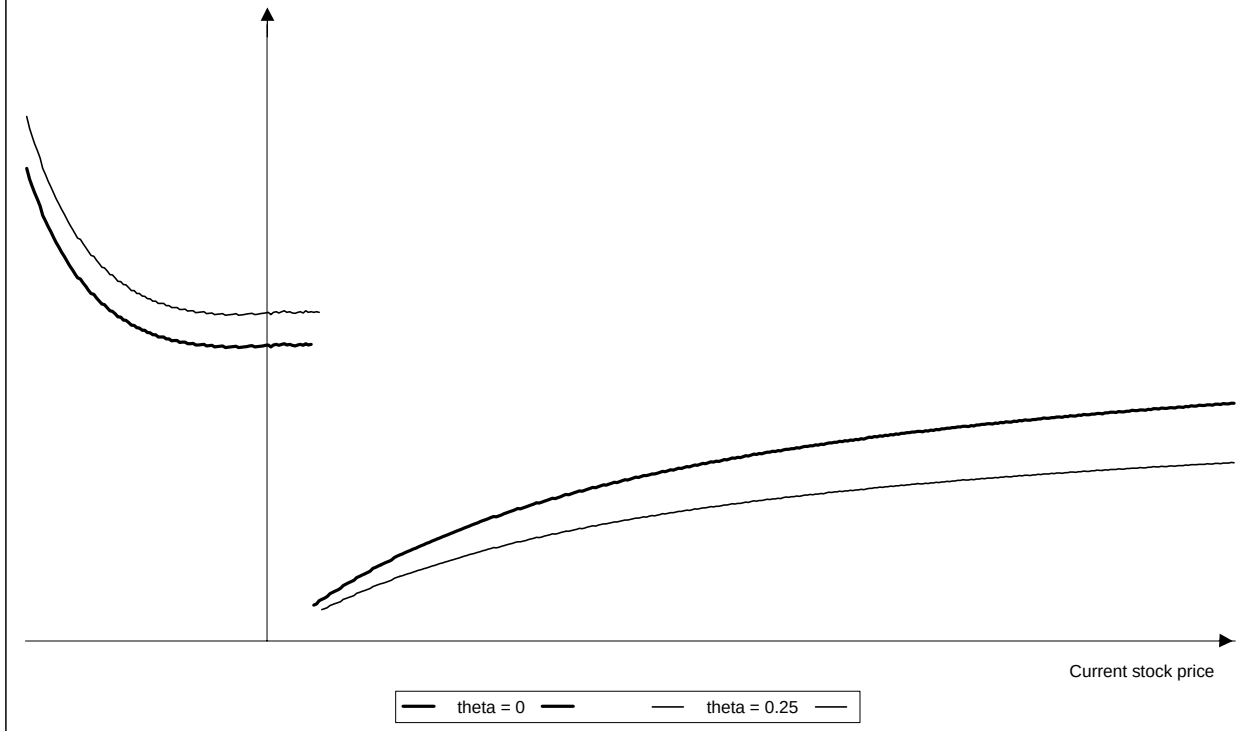


Figure 3a – Demand curve for the loss-averse investors in the two-period model of section 3. This figure represents the case in which the initial surplus wealth of the loss-averse investors is relatively “low”, and therefore they are not following the GPI strategy. The solid line plots the first-period demand curve and the initial allocation corresponds to point 0. The other two curves plot the second-period demand curves and allocations for the different possible shocks, positive (1b) or negative (1a).

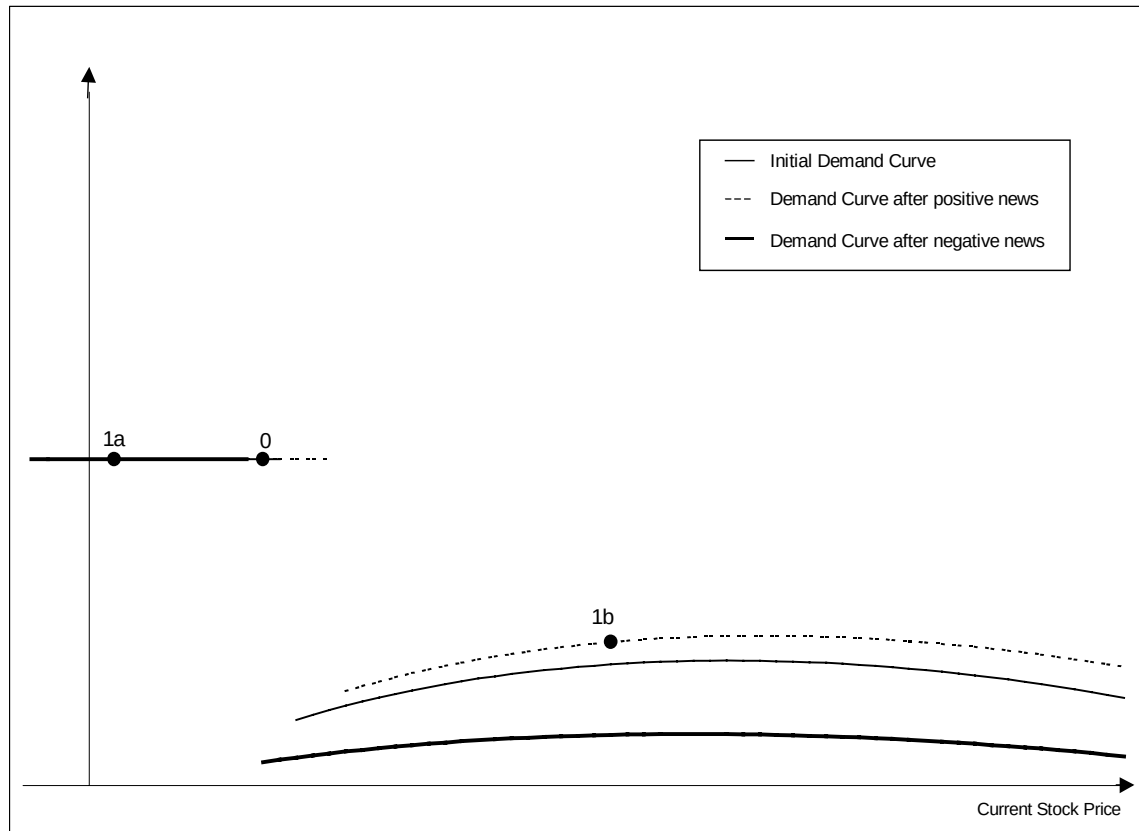


Figure 3b – Gross Stock Return in the good state, when the loss-averse investor’s initial surplus wealth was “low” (from 0 to 1b in figure 3a). Results are shown for different values of the news shock (Ω_h), and different values of the ratio of wealth between the CRRA investors and the loss-averse investors (WC/WLA).

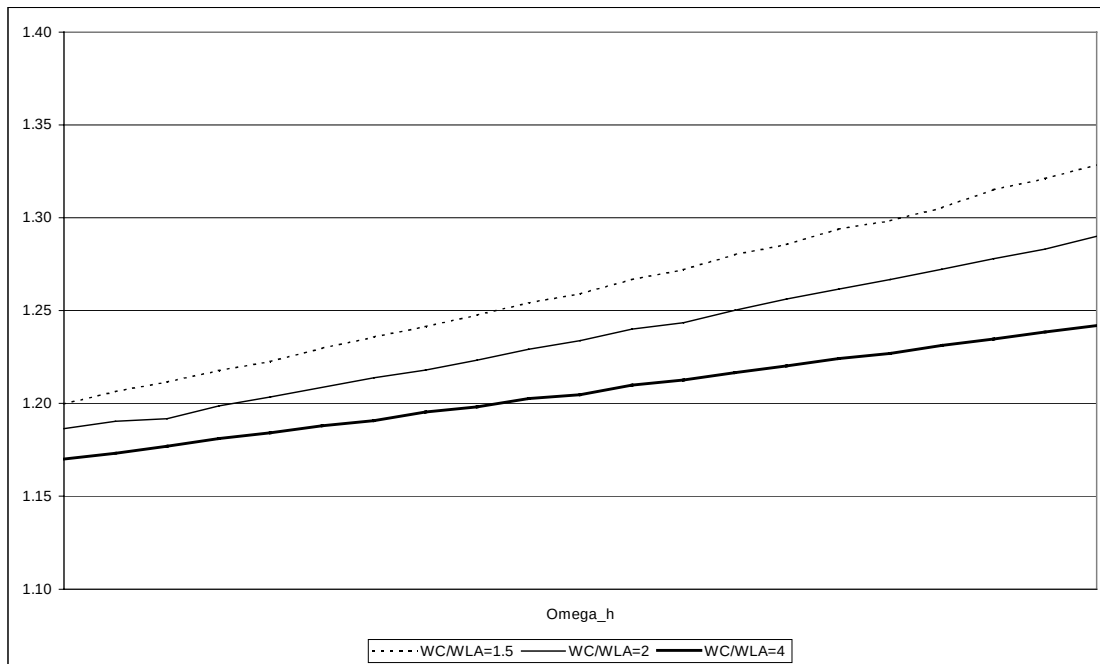


Figure 3c – Turnover Ratio in the good state, when the loss-averse investor’s initial surplus wealth was “low” (from 0 to 1b in figure 3a). Results are shown for different values of the news shock (Ω_h), and different values of the ratio of wealth between the CRRA investors and the loss-averse investors (WC/WLA).

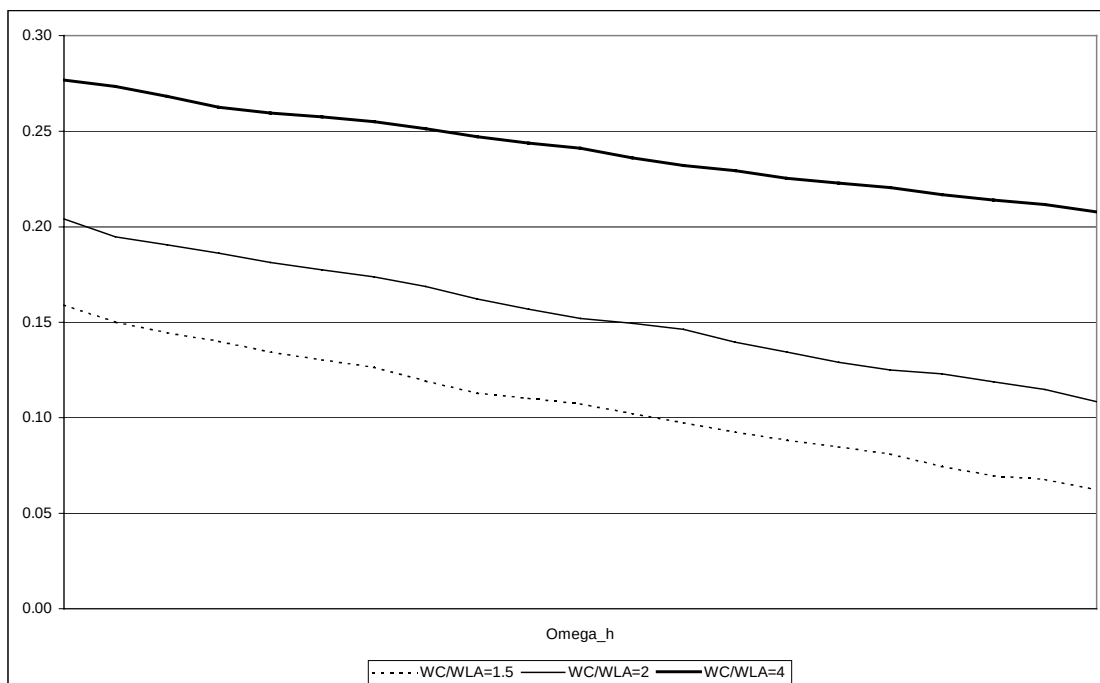


Figure 3d – Gross Stock Return in the good state, when the loss-averse investor’s initial surplus wealth was “low” (from 0 to 1b in figure 3a). Results are shown for different values of the news shock (Ω_h), and different rates of adjustment for the reference point.

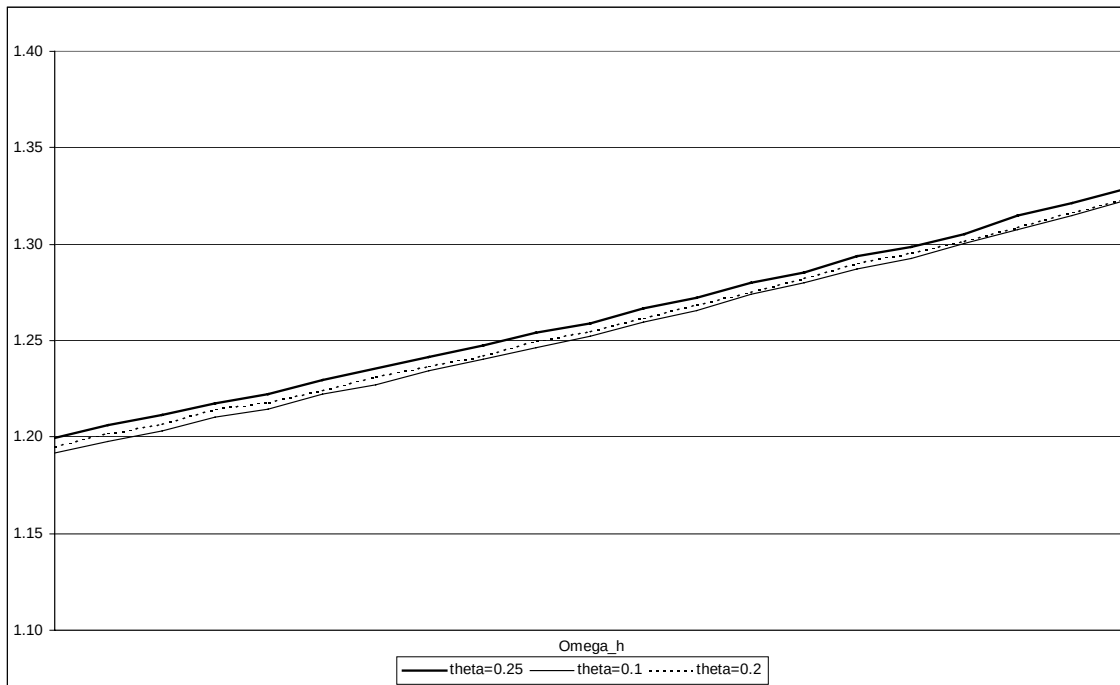


Figure 3e – Turnover Ratio in the good state, when the loss-averse investor’s initial surplus wealth was “low” (from 0 to 1b in figure 3a). Results are shown for different values of the news shock (Ω_h), and different rates of adjustment for the reference point.

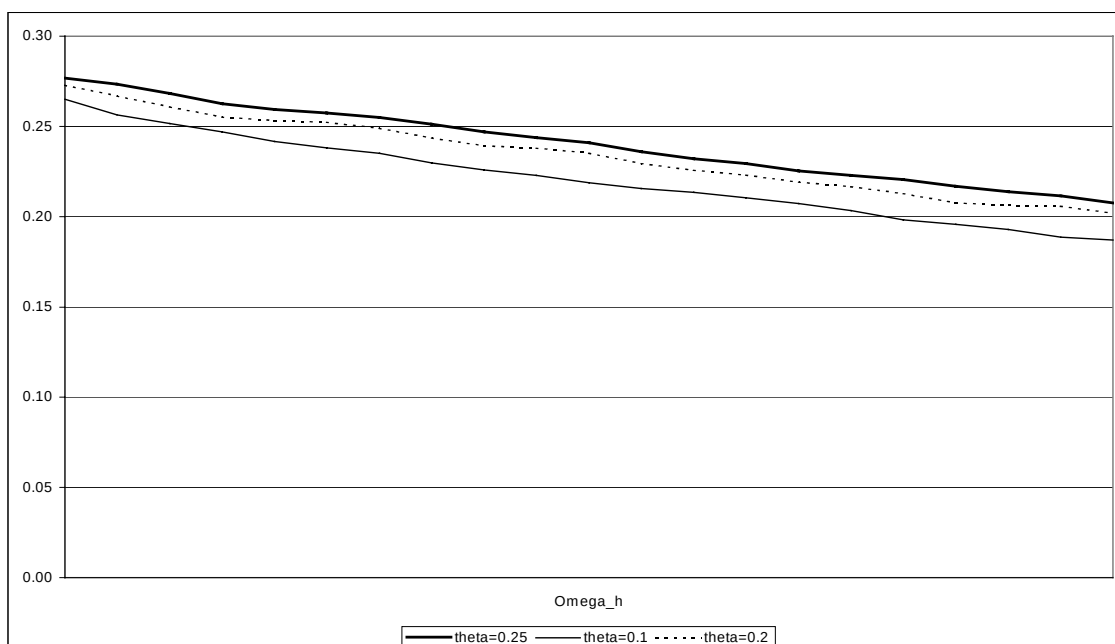


Figure 4a – Demand curve for the loss-averse investors in the two-period model of section 3. This figure represents the case in which the initial surplus wealth of the loss-averse investors is relatively “high”, and therefore they are following the GPI strategy. The solid line plots the first-period demand curve and the initial allocation corresponds to point 0. The other two curves plot the second-period demand curves and allocations for the different possible shocks, positive (1b) or negative (1a).

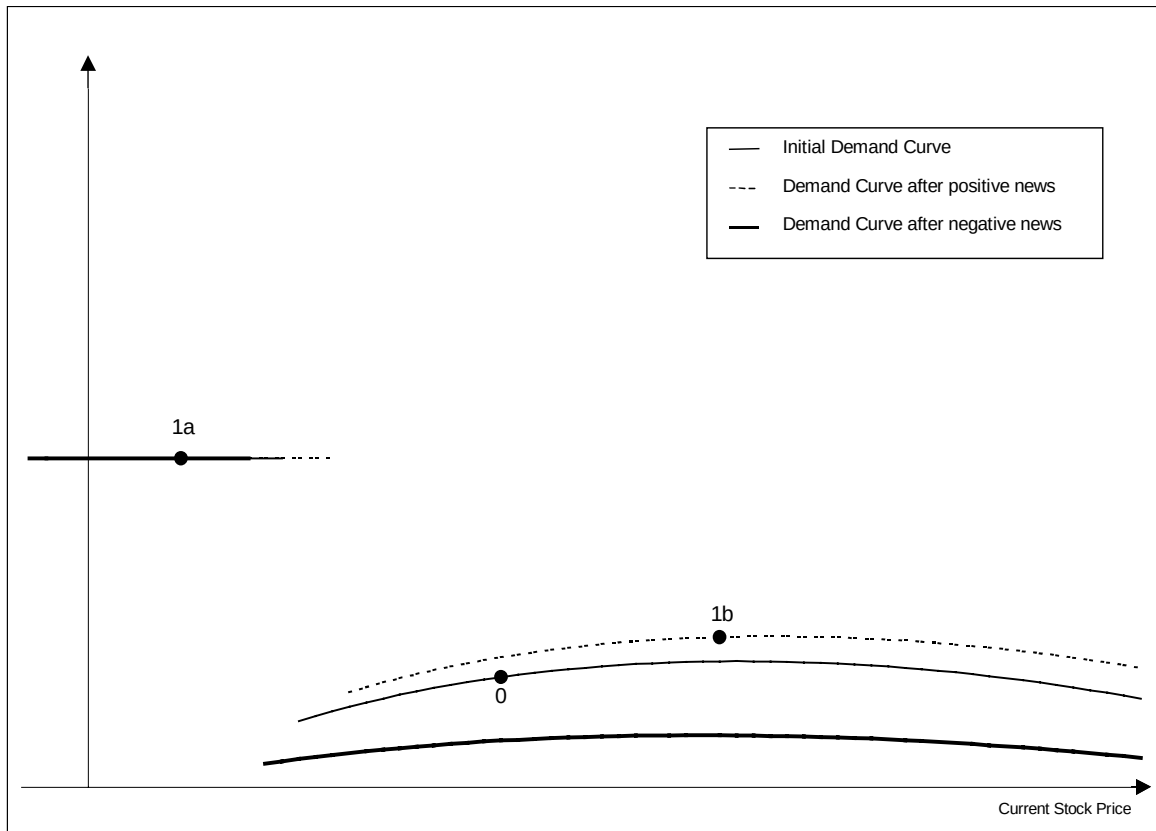


Figure 4b – Turnover Ratio in the good state, when the loss-averse investor’s initial surplus wealth was “high” (from 0 to 1b in figure 4a). Results are shown for different values of the news shock (Ω_h), and different values of initial surplus wealth for loss-averse investors, as fraction of their total wealth (SW).

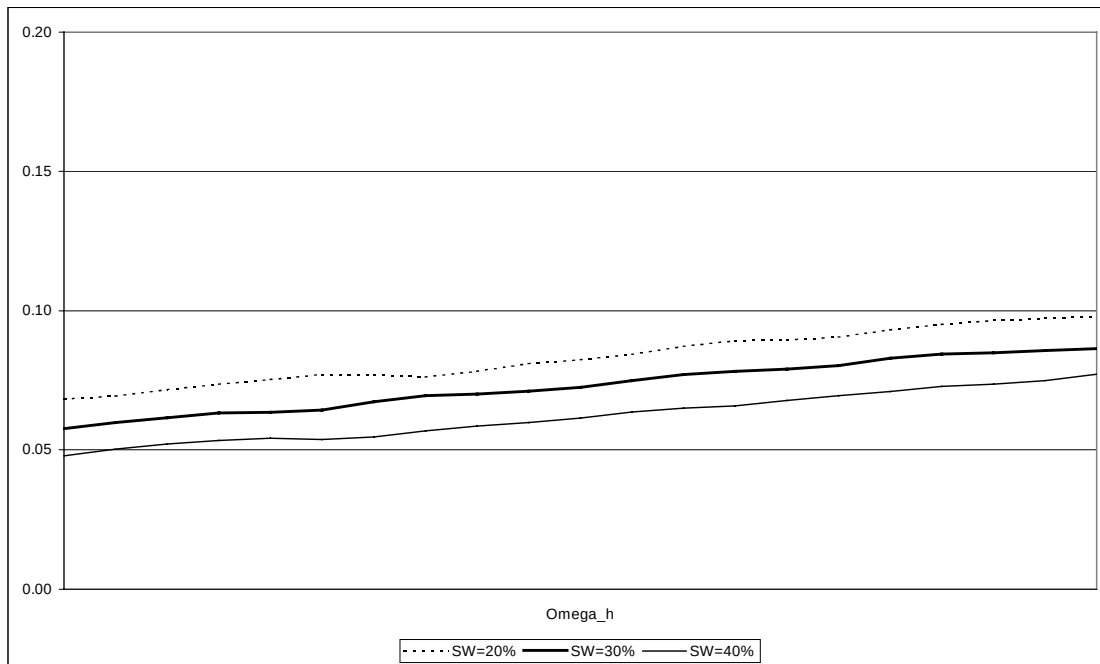


Figure 4c – Turnover Ratio in the bad state, when the loss-averse investor’s initial surplus wealth was “high” (from 0 to 1b in figure 4a). Results are shown for different values of the news shock (Ω_l), and different values of initial surplus wealth for loss-averse investors, as fraction of their total wealth (SW).

